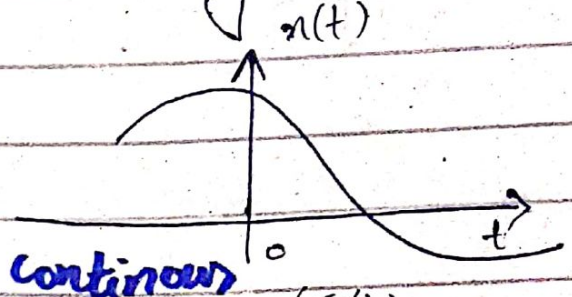
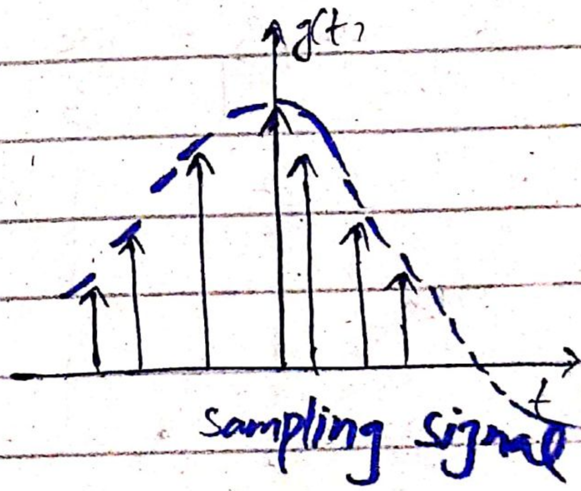
Sound

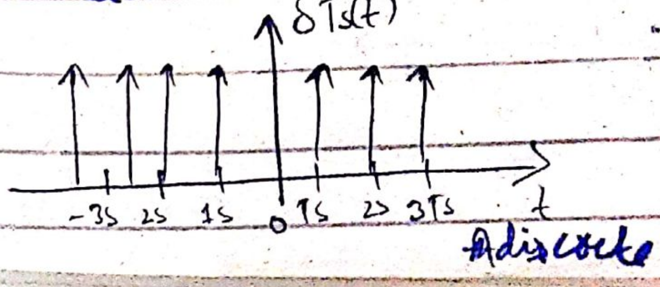
What is Sampling?
What is upsampling?
What is downsampling?

Sampling:

- it is the process of transforming a musical source into a digital file.
- it is the reduction of a continuous-time signal to a discrete-time signal.
- A sample is the value of signal at a point.
- Sampling theorem help us to convert continuous time signal to discrete time signal.







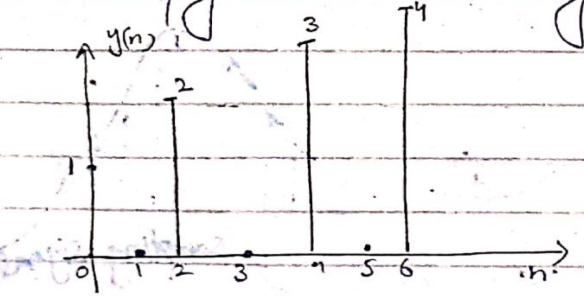
Up Sampling:

- By up sampling, sampling rate of the signal is increased.
- The quality of signal is increased through upsampling.

$F_x \longrightarrow \boxed{\uparrow I} \longrightarrow F_y$

$F_x = I F_y$

- The distance b/w two samples is called sampling rate. it is denoted by T.
- By increasing the sampling rate means the distance b/w the samples is decreasing.
- we stuff further information between that information that is currently transfer.
- Simply we are inserting sample.



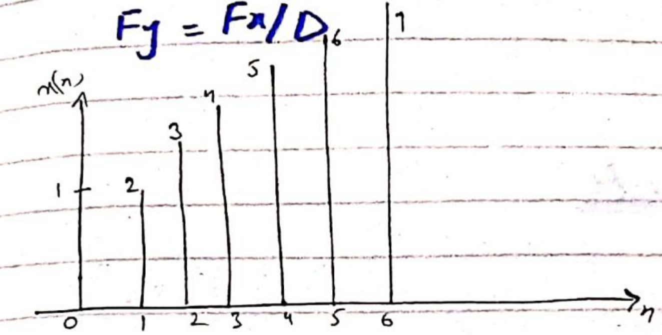
Down Sampling

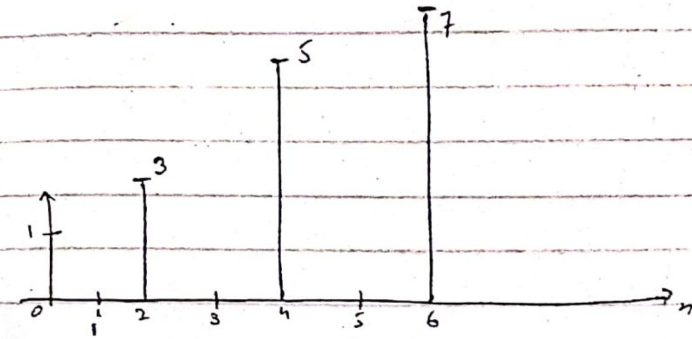
- In down sampling, we reduce the sampling rate, so the distance b/w two samples will increase and as a result we will drop the sample.

$F_x \longrightarrow \boxed{\downarrow} \longrightarrow F_y$

$F_x > F_y$

$F_y = F_x / D$





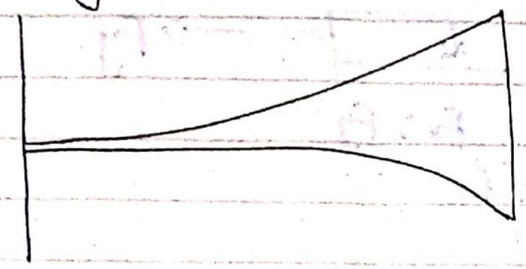
$F_y > F_x$

What is Fade ins & Fade out?

Fading is commonly used in audio transition.

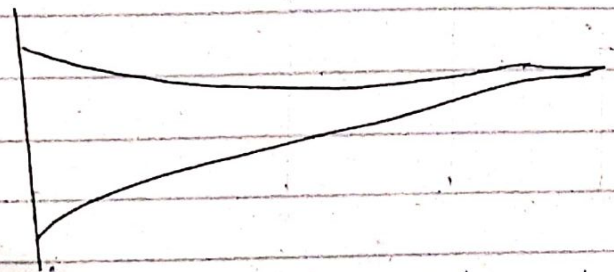
Fade-in:

=> Through fade in, the sound begins with silence and increase to the full volume gradually.



Fade-out:

=> Through fade out, the audio begins at full volume and decreases to silence gradually.



Both lets dissolve into and out of any object by ramping the opacity of the object from 0 to 100 percent and then back to 0 percent at end.

What is audio-equalization?

=> it is the process of adjusting the balance b/w frequencies within an audio signal.

=> This process increases or decreases the relative amplitudes of some frequency bands compared to other bands with filters, boosts and cuts.

=> it is used in audio mixing, tone shaping, crossovers, feedback control etc.

What is time stretching?

=> Time stretching enables us to shorten or lengthen audio without changing its pitch.

=> we change the speed or duration of an audio signal without affecting its pitch.

=> Pitch scaling is the opposite of time stretching.

What is reverse sound?

=> Reverse tape effects are special effects created by recording sound onto magnetic tape and then physically reversing the tape so that when the tape is played back, the sounds recorded on it are heard in reverse.

- Back masking is a type of reverse tape effect.

Describe different Components & measurements of sound?

⇒ The differences between sounds are caused by:

- intensity
- pitch
- tone.

Intensity:

- is the amount of energy a sound has over an area.
- The same sound is more intense if we hear it in a smaller area.
- Sounds with higher intensity are louder.
- An amplitude increase, intensity also increases.

Amplitude:

- Strength of sound waves, which we perceive as loudness or volume.
- is a measure of energy.
- more energy a wave has, higher its amplitude.

- Sounds are measured in decibels dB.

$$\Delta I\,(db) = 10 \log \frac{I_2}{I_1}$$

Pitch: — Position of a single sound in the complete range of sound.

- helps us distinguish between low and high sounds.
- Pitch depends on the frequency of a sound wave.

Frequency:

- is the number of wavelengths that fit into one unit of time.
- measured in hertz.

Ultrasonic:

- Sounds that are too high for us to hear are called ultrasonic.

Acoustics:

- is the branch of Physics that studies sound.
- Sound pressure levels are measured in decibels.
- a decibel measurement is actually the ratio b/w a chosen reference point on a logarithmic scale and the level that is actually experienced.

Q. What is digtal audio?

=> Digital audio data is the actual representation of sound stored in the form of "Samples."

=> Samples represent the amplitude (or loudness) of sound at a discrete point in time.

=> The quality of digital recording depends on the sampling rate.

=> Microphone produces analog signal.

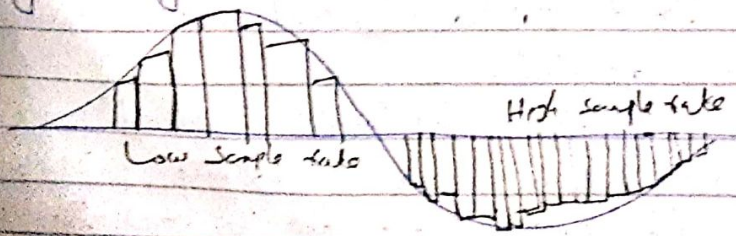
=> Computer understand only discrete entities.

=> Therefore analog audio is converted into digital audio. Also known as sampling.

Q. What is sampling rate?

=> The number of samples taken per second is called sampling rate.

=> Sampling rate can also be define as the number of samples per second taken from a continuous signal to make a discrete or digital signal.



Q What is sample size?

=> The number of bits used to describe the amplitude of a sound wave. or amount of info is stored.

Q. What sampling frequencies are most commonly used in multi-media?

Three sampling frequencies most often used in multi-media are CD-quality 44kHz, 22.05 kHz and 11.025 kHz.

Q. What are the crucial aspects of preparing digital audio files?

=> Crucial aspects of preparing digital audio files are:

- Balancing the need for sound quality against available RAM and hard disk resources.
- Setting appropriate recording levels to get a high quality and clean recording.

Q. What are basic editing operations?

- Trimming
- Splicing
- Assembly
- Volume adjustments
- Working on multiple tracks.

=> Additional available sound editing operations are:

=> format conversion
=> resampling or downsampling
=> fade ins and fade-outs
=> Equalization
=> Time stretching
=> digital signal processing
=> Reversing sounds

Trimming:

- Removing "dead air" or blank space from the front of a recording and any unnecessary extra time off the end.
- Trimming might be a few seconds and might make a big difference in your file size.
- it can be done by dragging the mouse cursor or by choosing a menu command such as cut, clear, Erase or Silence.

Splicing & assembly:

=> Removal of extraneous (impertinent) noises that inevitably creep into a recording and touch up is called Splicing. it is done by Adobe Premiere or final cut or by splices manually.
=> Creating a longer recording by mixing many small recordings is called assembly.
=> Same tools used that are used for trimming.

Splicing - to join two or more separate pieces of video together into a continuous sequence.
- it involves cutting and then physically attaching the ends of two or more pieces.

Volume adjustments:-

=> When we combine one or more recording with different volume level, they must be made to run with a consistent vol. level.
=> we need to normalize it with a sound editor to a particular level.
=> Don't increase the volume too much, this might distort the file.

Format-Conversion:

=> While Saving different formats are made available for sounds. Data may be lost during Conversion. MP3, MP4.

Equalization:

=> Equalization allow us to modify a recordings frequency content so that it sounds brighter (more high frequencies) or darker (low, ominous rumbles). Done to compensate the differences of the original recording

Time Stretching:

=> it allows us to alter the length of the file without changing its pitch.
=> it is useful but degrade the quality of the file.

Digital Sound Processing:

=> Allows signals with reverberation multitape delay, chorus, flange and other special effects using DSP routines.

Reversing Sounds:

=> it means reversing all or a portion of a digital audio recording.

Multiple-Tracks:

=> Being able to edit and combine multiple tracks, merge tracks and then export them in a final mix to a single audio file is important.

What are types of digital audio?

=> There are two types of digital audio:

- Stereo
- Monophonic

Stereo:

- when a sound comes from two different sources, it is called stereo audio.

formula:

stereo = Sampling rate * duration of recording in Seconds * (bit resolution/8) * 2

=> The number of channels is 2 for stereo.

Monophonic:

=> When a sound comes from a single source, it is called monophonic audio.

formula:

mp = Sampling rate * duration of recording in seconds * (bit resolution/8) * 1

=> The number of channels for monophonic is 1.

What is audio resolution?

=> Audio resolution determines the accuracy with which a sound can be digitized.

=> Audio resolution can be 8-bit or 16-bit.

=> Using more bits for the sample size yields a recording that sounds more like its original.

What is midi audio?

=> MIDI stands for musical instrument digital interface.

=> is a Communication Standard developed in the early 1980's for electronic musical instruments and computers.

=> it allows music and sound synthesizers from different manufacturers to Communicate with each other by sending messages along cables connected to the devices.

⇒ midi is a shorthand representation of music stored in numeric form.

⇒ it is not digitized sound.

⇒ A sequencer software and sound synthesizer is required in order to create MIDI scores.

⇒ MIDI is device dependent.

Q. What is difference b/w midi vs digital audio?

MIDI	Digital audio
○ Analogous it is analogous to structured or vector graphics.	○ it is analogous to bitmapped images
○ it is device dependent	○ it is device independent.
○ MIDI files are much smaller 200 to 1000 times	○ Digital files are not smaller.
○ MIDI files sound better when played on a high quality midi devices.	○ Digital audio files sound is not better.
○ it is difficult to play back spoken dialog with midi.	○ while digitized audio can do so with ease.
- Never records human voice.	can record human voice.
- Not contain a recording of sound.	- Contain a recording of sound
○ it does not have consistent playback quality.	○ Digital audio provides a consistent playback quality.
○ In order to run MIDI, one requires knowledge of music theory.	○ while digital audio does not have this requirement.
Better when played on high quality midi device	quality is not so better

Com

Q. What is audio file format? and discuss its different types?

Audio-file format:-

⇒ Audio is an electrical or other representation of sound. Audio format defines the quality and loss of audio data.

⇒ An audio file format is a file format for storing digital audio data on a computer system.

⇒ it can be a raw bitstream, but it is usually a container format or an audio data format with defined storage layer.

(i) MP3:

⇒ MPEG-1 Audio layer 3(AC3) file.

⇒ MP3 is a standard technology and format for compressing a sound sequence into a very small file (about one-twelfth the size of the original file) while preserving the original level of sound quality when it is played.

=> MP3 provides near CD quality audio.

=> it is a lossy compression.

=> Main aim of MP3 is to remove all those sounds which not hearable or less noticeable by human ears.

=> MP3 is like universal format which is Compatible almost every device.

(2) WAV:- Developed by microsoft and IBM in 1991

=> Stands for Wave form audio file format.

=> Wav is a file extension for an audio file format Created by microsoft

=> The wav file has become a standard PC audio file format for everything from system and game sounds to CD-quality audio. it is just a windows containes for audio formats.

=> Also referred to as plus code modulation (PCM). it is just a wrapper for the PCM encoding
- Compatible with windows & MAC

=> A wav file is uncompressed audio.

=> A wave file also stores information about the file's number of tracks, sample rate, bit depth and whether it's mono or stereo.

(3) WMA: – 1999

=> Short for window media Audio, WMA is a microsoft file format for encoding digital audio files similar to MP3 though can compress files at a higher rate than MP3.

=> it was designed to remove some of the flaws of MP3 compression method.

=> In terms of quality it is better than MP3. => it is least efficient in term of compression and is not open-source

=> But it is not most widely used. it has limited hardware support.

(4) AAC:

- Stands for advanced audio coding.

=> The compression algorithm used by AAC is much more complex and advanced than MP3. Developed in 1997 after MP3.

=> AAC have better sound quality as compared to other particular audio file in MP3.

=> it is the standard audio compression method used by youtube, Android, iOS, itunes, and playstations.

5) Ogg (.ogg):

=> Ogg is an audio compression format comparable to other formats used to store and play digital music, but differs in that it is free, open and unpatented.

=> Ogg refers to the container format.

=> it was created by Xiph.Org.

6) AIFF:

=> it stands for audio interchange file format.

=> it was developed by Apple for mac systems.

=> Like WAV files, AIFF files can contain multiple kinds of audio.

=> it contain uncompressed audio in PCM format

=> it just a wrapper for the PCM encoding.

=> it is compatible with both mac and windows.

What is sound synthesizer?

=> A sound synthesizer is an electronic instrument that uses some form of digital or analog processing to produce audible sound

=> The process of creating sounds using synthesizers is called synthesis

=> it electronically generates and modifies sounds, frequently with the use of a digital computer. A good synthesizer often has ~~an~~ microprocessor, keyboard, ~~controls~~ memory etc.

=> Synthesizers are used for the composition of electronic music. it has one or more MIDI in and out.

=> it is also called sampler

=> it can use a variety of synthesis or sample based synthesis to make sound.

What is Track?

=> Track is a sequencer is used to organize the recordings

=> Tracks can be turned on or off on recording or playing back.

What is Timbre?

=> The quality of the sound. e.g flute sound, cello sound etc.

=> multi-timbral: capable of playing many different sounds at the same time.

=> The control settings that define a

particular timbre.

(5) ## What is voice?

- Voice is the portion of the synthesizer that produces sound.
- Synthesisers can have many (12, 20, 24, 36) etc) voices.
- Each voice works independently and simultaneously to produce sounds of different timbre & pitch.

What is Sequencer Software?

It can be stand-alone hardware unit or a Software running on a Computer.

Audio formats are divided into Three parts:
- Uncompressed format (PCM, WAV, AIFF)
- Lossy Compressed format (MP3, AAC, WMA)
- Lossless Compressed format (FLAC, ALAC)

~~Q. How to add Sound in multi-media project?~~

=> MIDI is a connectivity standard that musicians used to hook together musical instruments and Computer equipment.

=> It is a technical standard that describes a Communication protocol, digital interface and electrical connector that connect various electronic musical instrument and related audio devices for playing, editing and recording music.

=> It is a way to connect devices that make and control sound - such as synthesizers, Samplers and Computers So that they can Communicate with each other, using midi message.

=> it is a description of how to Create the sound based on predefined sound. MIDI recording does not contains the human voice and does not recorded (transmit) sounds.

=> Midi files are basically text document So they take little interface.

- Parts of MIDI:

=> Physical connector message format storage format.

MIDI files may be converted to mp3, wav, WMA on any windows platform using total Audio.

Parts or Components:

- Synthesiser or Sampler
- Sequencer
- Computer MIDI control I/O and O/O

Lossless file format for audio:

FLAC

- It stands for free lossless audio codec. It can compress a source file by up to 50% without losing data. It is most popular in its category and is open-source.

ALAC

- It stands for Apple lossless audio codec. It was launched in 2004. And became free after 2011. It was developed by Apple.

PCM:

- PCM stands for pulse-code modulation. It represents raw analog audio signals in digital form.
  - It is most common format used in CDs and DVDs.
- It is uncompressed file format.

Sampling:

- The first step in converting an analog quantity to digital form is called sampling.
- In signal processing, sampling is the reduction of the continuous signal to a discrete signal.

e.g. Conversion of a sound wave into a sequence of samples.

Quantization:-

⇒ The amplitude values obtained after sampling may be long real numbers which are usually rounded to the nearest predefined discrete values.

⇒ This process of converting the real numbers to the predefined discrete numbers is called quantisation.

⇒ The discretization of the amplitude values is called quantization.

Animation tools

- What is Adobe's Flash?

- it is a multi-media Software platform used for production of animations.
- it was originally developed by FutureWave.
- Flash was first known as "FutureSplash Animator."
- it was a popular Software to Create digital Content containing elements like the following:
  - Animations
  - motion graphics
  - Streaming Video
  - interactivity
  - Audio
  - text
- Developers used these interactive features to create rich web applications, embedded web browser video players, desktop and mobile applications, mobile games and interactive video player.
- The file extension of Adobe Flash **.FLV.**
- it is now called "Adobe animate".

What is auto-desk's Maya?

- Autodesk maya, Commonly shortened to just maya, is a 3D Computer graphics application that runs on windows, mac OS and Linux, originally developed by Alias and currently owned and developed by Autodesk.
- Maya is an abbreviation of: "Most advanced. Yet Acceptable"
- it is used to create:
  - interactive 3D applications (Including ...)
  - video games
  - Animated films
  - TV series
  - Visual effects
- The maya API is a C++/Python API.
- Maya supports a wide range of file formats, Commonly used for save maya Scenes is **.ma.**

Q. What is NewTek's lightwave?

- Lightwave 3D is a 3D Computer graphic program developed by NewTek.
- It is Comprehensive, reasonably priced and well recognized 3D system for producing incredible

[illegible] windows and macOS.

- Lightwave 3D combines a state of the art renderer with powerful intuitive modeling and animation tools.
- It is an open source software.
- The latest version of NewTek Lightwave is 3D 2020.
- Many companies such as [illegible] used Lightwave 3D currently.

Q. What is Papervision 3D?

- It was developed to enable 3D graphics with flash.
- It uses actionscript as the language.
- It was developed by a core team and released to public in 2007 and became open source.
- Papervision allows to make 3D objects and manipulate them.
- Works by initializing a scene having a camera & object and then adding any other needed objects to the scene.
- An object position in papervision is defined by x, y & z and path.
- Papervision positions in space are depicted with three x, y, z coordinates.
  - x axis = width i.e. left (-) to right (+)
  - y axis = Height i.e. down (-) to upper (+)
  - z axis = Depth i.e. close / forward (+) to far / backward (-)

Q. What GreenSock's Tween max?

- Tweens are the basic building block of a greensock animation.
- A "Tween" is a single transition.
- The greensock animation platform is a powerful javascript library that enables front-end developers and designers to create robust timeline based animation.
- Tween animation calculates the animation with information such as the start point, end point, size, rotation and other common aspects of animation.
- Tween.method (element, duration, props)
- It is built for professionals.

Q. What is 3D Studio max?

- It is a professional 3D computer graphics program for making 3D animation, models, games.

- it has flexible plugin architecture and modeling capabilities and can be used on microsoft windows platform.
- Also known as 3D max.
- used for:
    - Character modeling & animation
    - Rendering photorealistics images

etc.

- The file extension is .3ds.

The software used in 2D animations is
Adobe photoshop
Adobe flash
After effects
Encore.

The software used in 3D animation is:

zBrush
Houdini
Blender
Autodesk maya.